

Project Documentation

Date	01 October 2026
Team ID	SWTID-2026-5325
Project Name	Import Data Using Transform Maps (Spreadsheets)
Maximum Marks	4 Marks

Project Description:

This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. The workflow loads spreadsheet data into an Import Set Table, maps and transforms the fields, validates the data, and stores the processed records in the Employee Test target table.

The spreadsheet data is first loaded into an Import Set Table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields. Employee ID is configured as the Coalesce field so that an existing employee record can be updated when the same Employee ID is imported again, while a new Employee ID can create a new target record.

Upon executing the transformation, records are created in the target table and validated to ensure data integrity. The project also includes Reports and a Dashboard to present the resulting employee data.

Scenarios

- Scenario 1: An HR/admin user prepares an employee spreadsheet and loads it into the ServiceNow Import Set Table for staging.

- Scenario 2: The Transform Map maps employee fields, validates the imported values, and uses Employee ID as the Coalesce field to update matching records rather than creating another copy.
- Scenario 3: The team imports new employee information, verifies the target records, and reviews the results through Reports and the Dashboard.

Technical Architecture:

Description: The solution uses a staged import architecture. Employee data begins in an Excel spreadsheet, is loaded into an Import Set Table, and then passes through a Transform Map for field mapping, transformation, and validation. Employee ID is used as the Coalesce key. The transformed data is stored in the Employee Test target table and is then surfaced through Reports and the Dashboard.

The project does not use a separate application database or standalone web server. The core data-processing and reporting activities are performed within the ServiceNow development instance.

Pre-requisites:

- ServiceNow development instance: Access to configure tables, Import Sets, Transform Maps, and Reports.
- Employee spreadsheet: Excel file containing the employee data to be imported.
- Import Sets knowledge: Basic understanding of staging tables and data loading.
- Transform Maps knowledge: Basic understanding of source/target field mapping and transformation.
- Data validation and Coalesce: Ability to validate imported records and configure Employee ID as the matching field.
- Reports and Dashboard: Access to create and review project result reports.

Project Workflow:

Milestone 1: Creation Of Spreadsheet And Table

- Activity 1.1: Create the employee spreadsheet with the required columns and values.
- Activity 1.2: Create the required ServiceNow tables and target structure for employee data.

Milestone 2: Creation Of importset Table And Transform Map

- Activity 2.1: Create the Import Set Table and load the spreadsheet data into the staging area.
- Activity 2.2: Create the Transform Map and define source-to-target field mappings.

Milestone 3: Transform Data & Validate And Enable Coalesce to Avoid Duplicate Records

- Activity 3.1: Transform and validate the imported data.
- Activity 3.2: Enable Coalesce on Employee ID to avoid duplicate records and update existing matches.
- Activity 3.3: Insert new employee data in Excel format and run the import workflow.
- Activity 3.4: Verify the Employee Test records and confirm the import result.

Milestone 4: Creation Of reports & dashboards

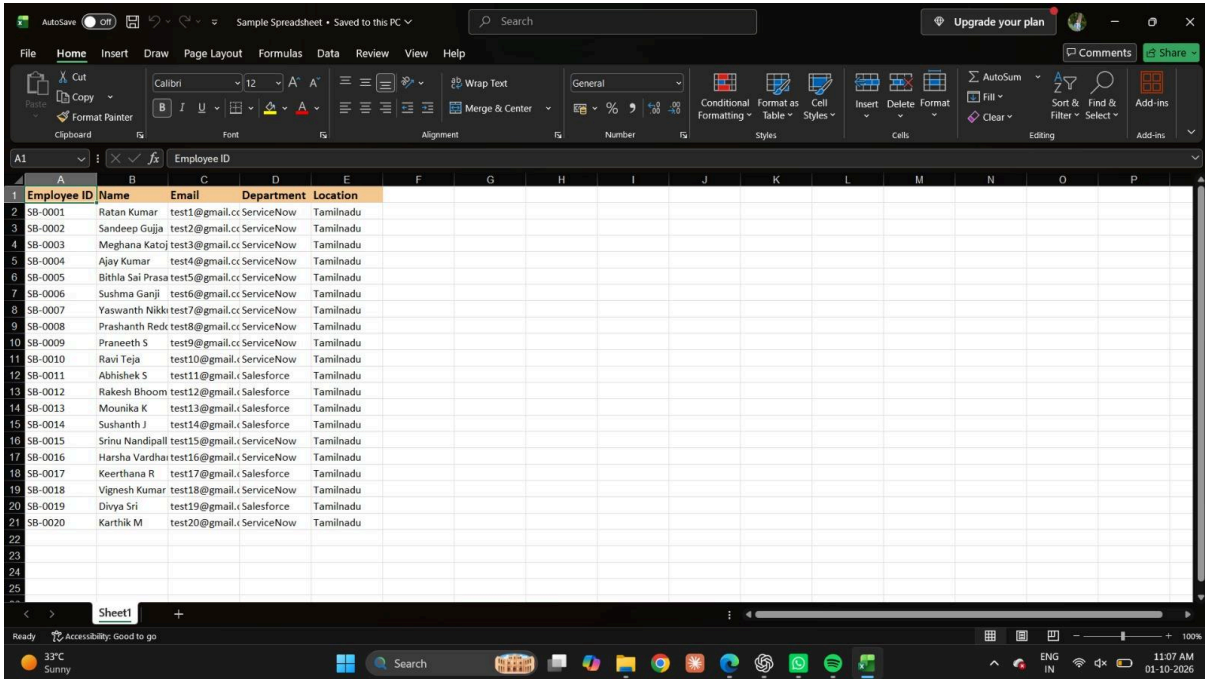
- Activity 4.1: Create reports for the imported employee data.
- Activity 4.2: Add the reports to the project Dashboard

Milestone 1: Creation Of Spreadsheet And Table

This milestone establishes the employee spreadsheet and ServiceNow table structure needed for the import workflow. The spreadsheet contains Employee ID, Name, Email, Department, and Location, and the target structure is prepared before transformation. The employee data must be organized consistently so that the later Import Set and Transform Map stages can process it reliably.

Activity 1.1: Creation of Spreadsheet

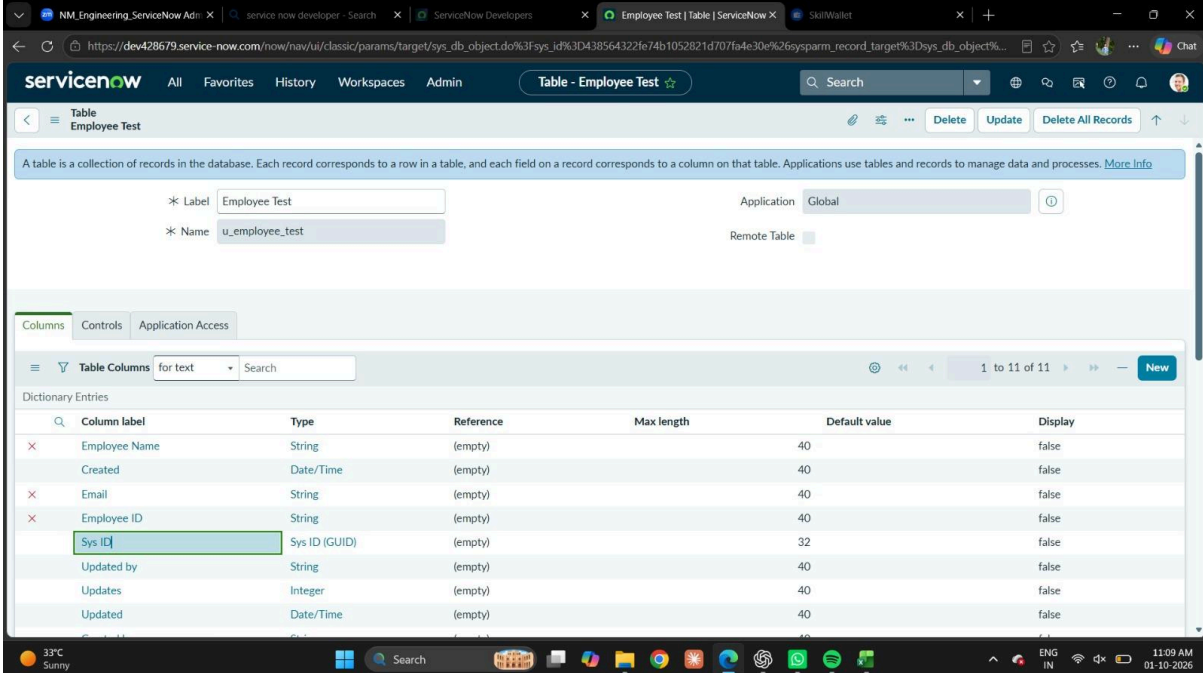
- Prepare employee data: Create the Excel spreadsheet containing employee information required by the project.
- Use consistent headers: Keep the spreadsheet column names aligned with the fields that will be mapped in ServiceNow.
- Provide Employee ID: Include Employee ID as the unique identifier used later for Coalesce matching.
- Review source values: Check the spreadsheet for missing, inconsistent, or incorrectly formatted values before loading it.



Employee ID	Name	Email	Department	Location
SB-0001	Ratan Kumar	test1@gmail.cc	ServiceNow	Tamilnadu
SB-0002	Sandeep Guja	test2@gmail.cc	ServiceNow	Tamilnadu
SB-0003	Meghana Katoj	test3@gmail.cc	ServiceNow	Tamilnadu
SB-0004	Ajay Kumar	test4@gmail.cc	ServiceNow	Tamilnadu
SB-0005	Bithla Sai Prasa	test5@gmail.cc	ServiceNow	Tamilnadu
SB-0006	Sushma Ganji	test6@gmail.cc	ServiceNow	Tamilnadu
SB-0007	Yaswanth Nikki	test7@gmail.cc	ServiceNow	Tamilnadu
SB-0008	Prashanth Redk	test8@gmail.cc	ServiceNow	Tamilnadu
SB-0009	Praneeth S	test9@gmail.cc	ServiceNow	Tamilnadu
SB-0010	Ravi Teja	test10@gmail.cc	ServiceNow	Tamilnadu
SB-0011	Abhishek S	test11@gmail.cc	Salesforce	Tamilnadu
SB-0012	Rakesh Bhoom	test12@gmail.cc	Salesforce	Tamilnadu
SB-0013	Mounika K	test13@gmail.cc	Salesforce	Tamilnadu
SB-0014	Sushanth J	test14@gmail.cc	Salesforce	Tamilnadu
SB-0015	Srinu Nandipall	test15@gmail.cc	ServiceNow	Tamilnadu
SB-0016	Harsha Vardhai	test16@gmail.cc	ServiceNow	Tamilnadu
SB-0017	Keerthana R	test17@gmail.cc	Salesforce	Tamilnadu
SB-0018	Vignesh Kumar	test18@gmail.cc	ServiceNow	Tamilnadu
SB-0019	Divya Sri	test19@gmail.cc	Salesforce	Tamilnadu
SB-0020	Karthik M	test20@gmail.cc	ServiceNow	Tamilnadu

Activity 1.2: Creation of Tables

- **Create target structure:** Create the ServiceNow table used for the employee records, including the Employee Test target table used by the project.
- **Define fields:** Create the fields required to store Employee ID and the other employee attributes contained in the spreadsheet.
- **Confirm field compatibility:** Ensure the source values can be mapped to the corresponding target fields.



The screenshot shows the ServiceNow interface for configuring a table named 'Employee Test'. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', 'Admin', and a search bar. The table configuration section includes fields for 'Label' (Employee Test), 'Name' (u_employee_test), 'Application' (Global), and 'Remote Table'. Below this, the 'Columns' tab is active, showing a list of columns with their labels, types, references, max lengths, default values, and display settings.

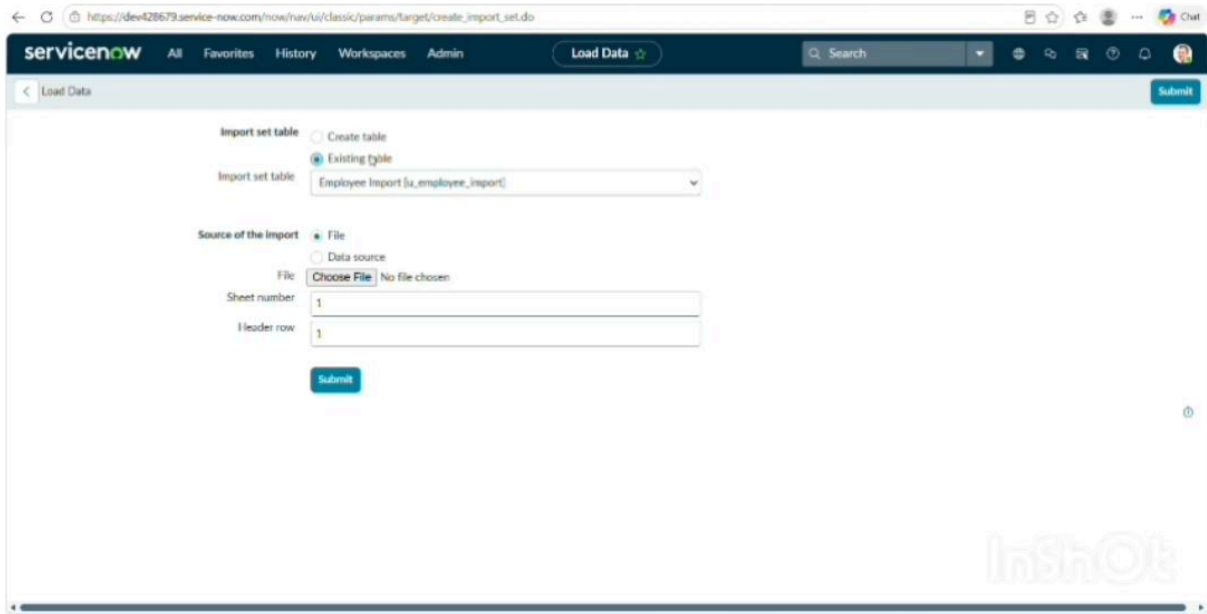
Column label	Type	Reference	Max length	Default value	Display
Employee Name	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Email	String	(empty)	40		false
Employee ID	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Updated	Date/Time	(empty)	40		false

Milestone 2: Creation Of importset Table And Transform Map

This milestone moves the spreadsheet into ServiceNow staging and defines the transformation rules used to populate the Employee Test target table.

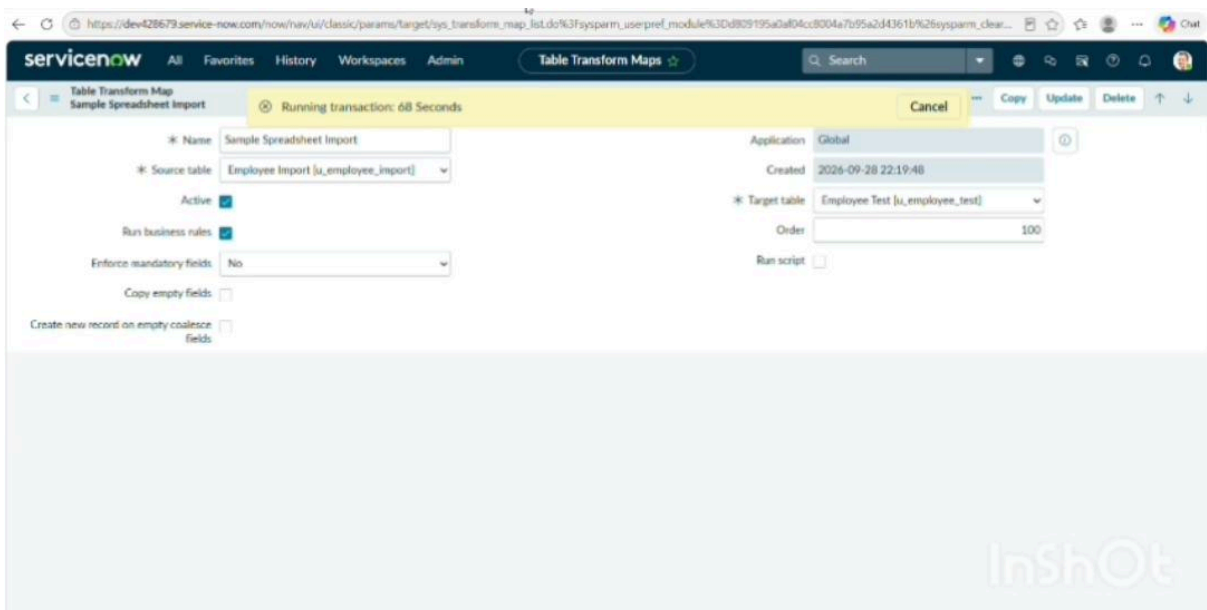
Activity 2.1: Create Import Set Table

- Load the spreadsheet: Use the ServiceNow import process to select the employee spreadsheet.
- Create/select the staging table: Use an Import Set Table as the temporary destination for the incoming rows.
- Review staged records: Confirm that the imported rows and source fields are visible before transformation.



Activity 2.2: Create Transform Map

- Create Transform Map: Create a Transform Map connecting the Import Set Table to the Employee Test target table.
- Define source and target: Select the staging table as the source and Employee Test as the target.
- Map fields: Create the source-to-target field mappings needed to transfer employee data.

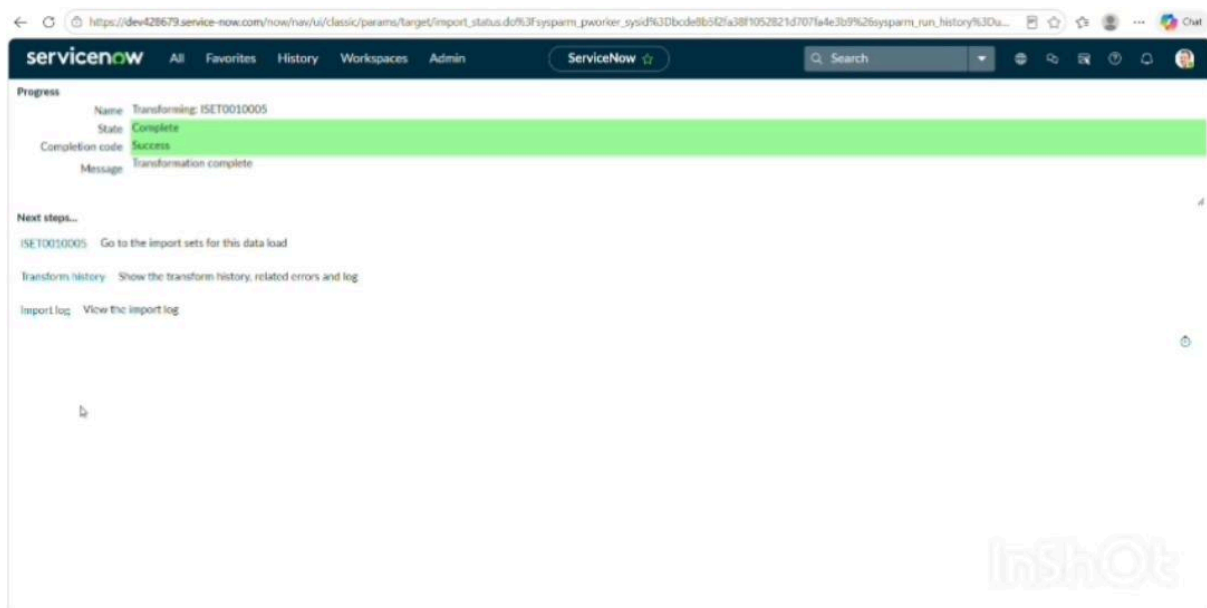


Milestone 3: Transform Data & Validate And Enable Coalesce to Avoid Duplicate Records

This milestone applies the Transform Map to the staged employee data and verifies that the transformation produces the expected target records.

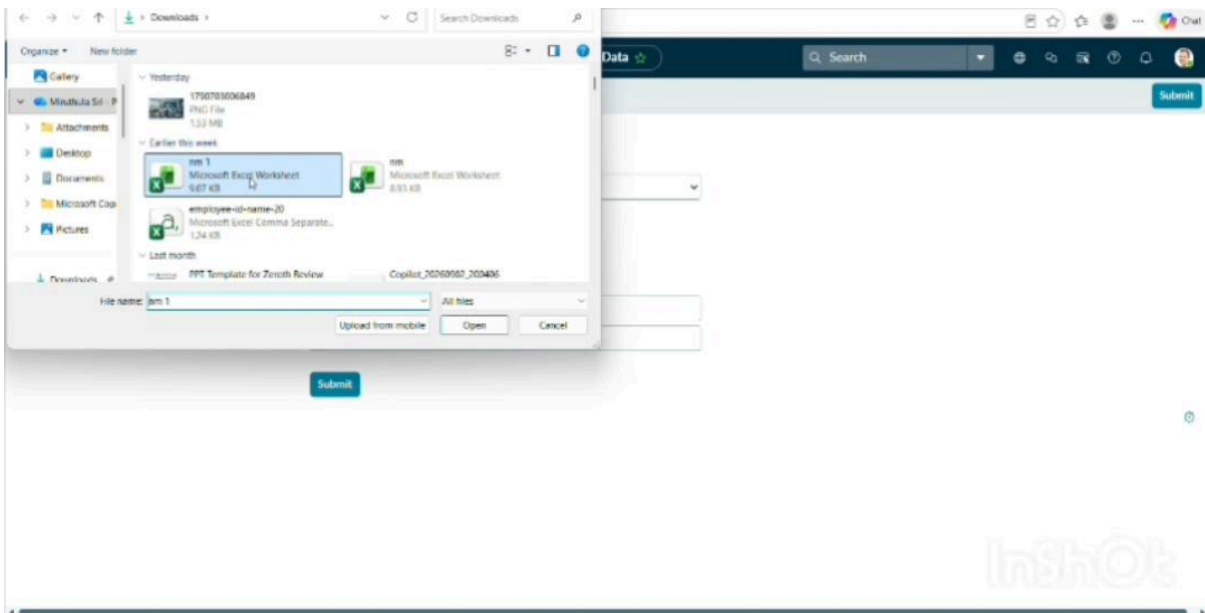
Activity 3.1: Transform Data and Validate

- Run the transformation: Execute the Transform Map against the staged employee records.
- Review field mapping: Confirm that source fields populate the intended Employee Test fields.
- Validate records: Check that the transformed records contain the expected Employee ID and employee information.
- Review the result: Use the import/transform result to identify successful processing or errors.



Activity 3.2: Enable Coalesce to Avoid Duplicate Data

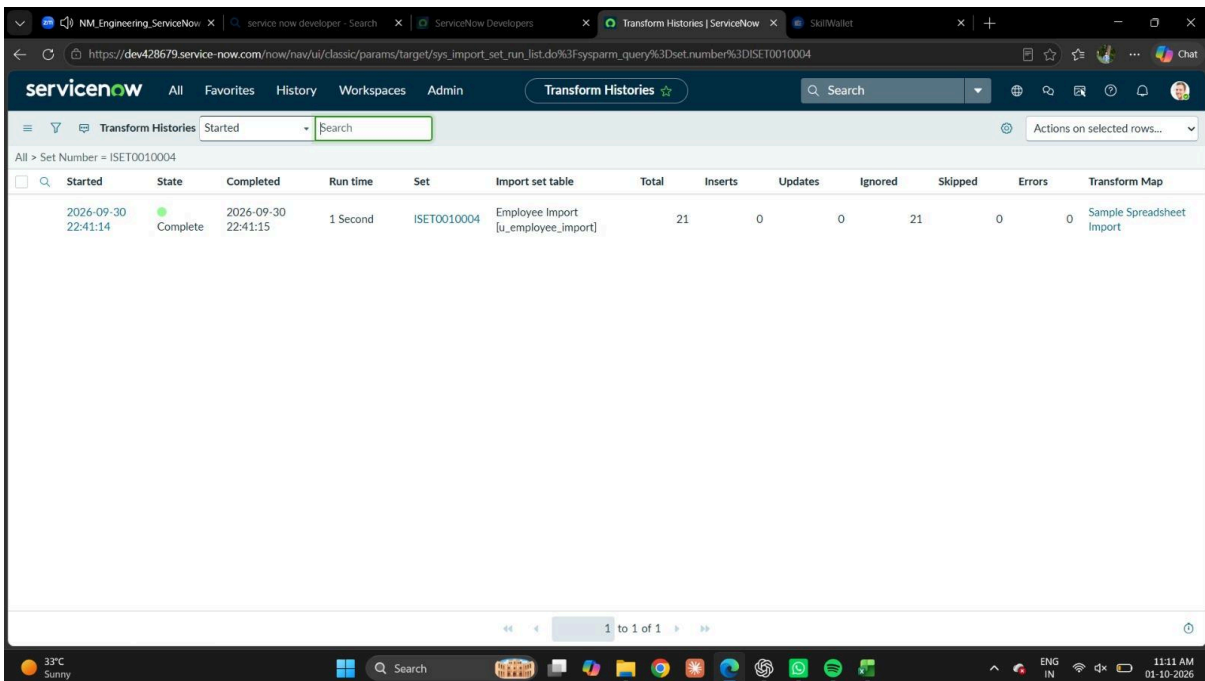
Employee ID is configured as the Coalesce field. In ServiceNow, a Coalesce field is used to match incoming source values against existing target records. When a match is found, the target record is updated; when no match is found, a new record can be inserted.



Coalesce is the data-integrity rule used to distinguish an existing employee from a new employee during transformation.

Activity 3.3: Transformation Result and Validation

The project validation results supplied for the workflow are summarized below. These values are project execution results provided for the documentation and are not inferred from a generic example.



Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors	Transform Map
2026-09-30 22:41:14	Complete	2026-09-30 22:41:15	1 Second	ISET0010004	Employee Import [u_employee_import]	21	0	0	21	0	0	Sample Spreadsheet Import

Validation Summary

- Records processed: 21 employee records.
- Import status: Success.
- Duplicate-record test: Yes.
- Existing-record update: Yes; existing records were updated correctly using Coalesce.
- Slow/error step observed: No.

The result should be supported in the final submission with the actual ServiceNow Import Set/Transform History and target-table screenshots if the evaluator requires visual evidence.

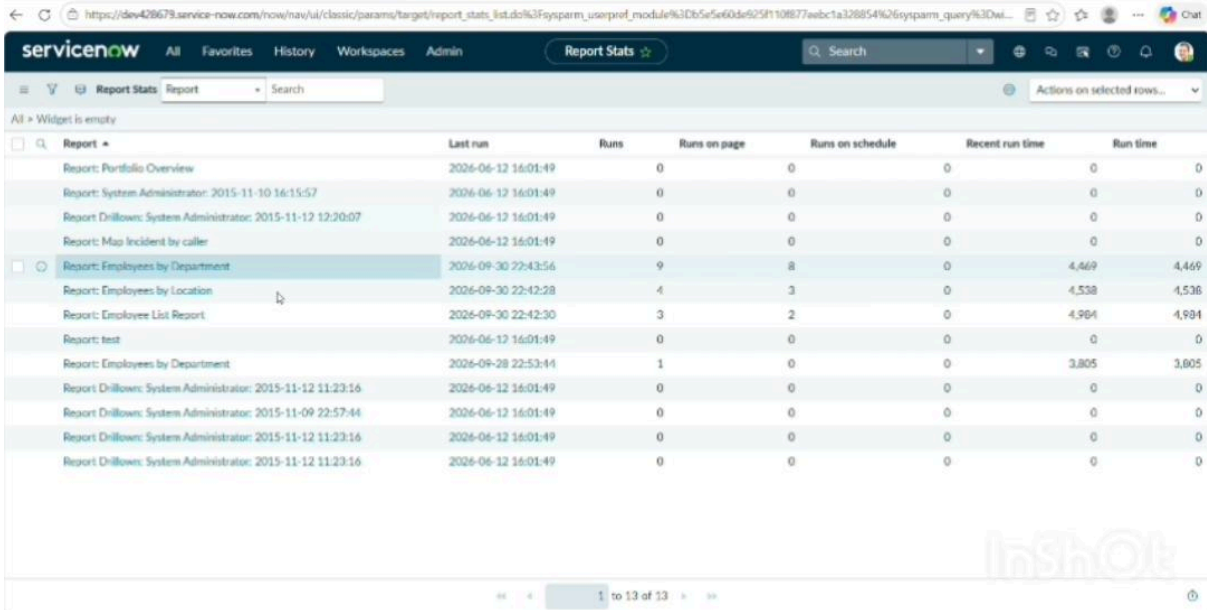
Milestone 4: Creation Of reports & dashboards

This milestone demonstrates the end-to-end use of the completed import configuration with new employee data in Excel format.

Activity 4.1: Create Reports – Detailed Steps

Create Reports on Employee Test Table

- Open All menus, Search for Reports, and under Usage and Governance select Reports.
- Click New to create the report. For Report 1, enter Report name: Employees by Department; Source Type: Table; Table: Employee Test.
- Click Next, select Pie chart, then configure Group by: Department and Aggregation: Count. Run the report and click Save.
- Create the remaining two reports using the same navigation: Employees by Location and Employee List Report.



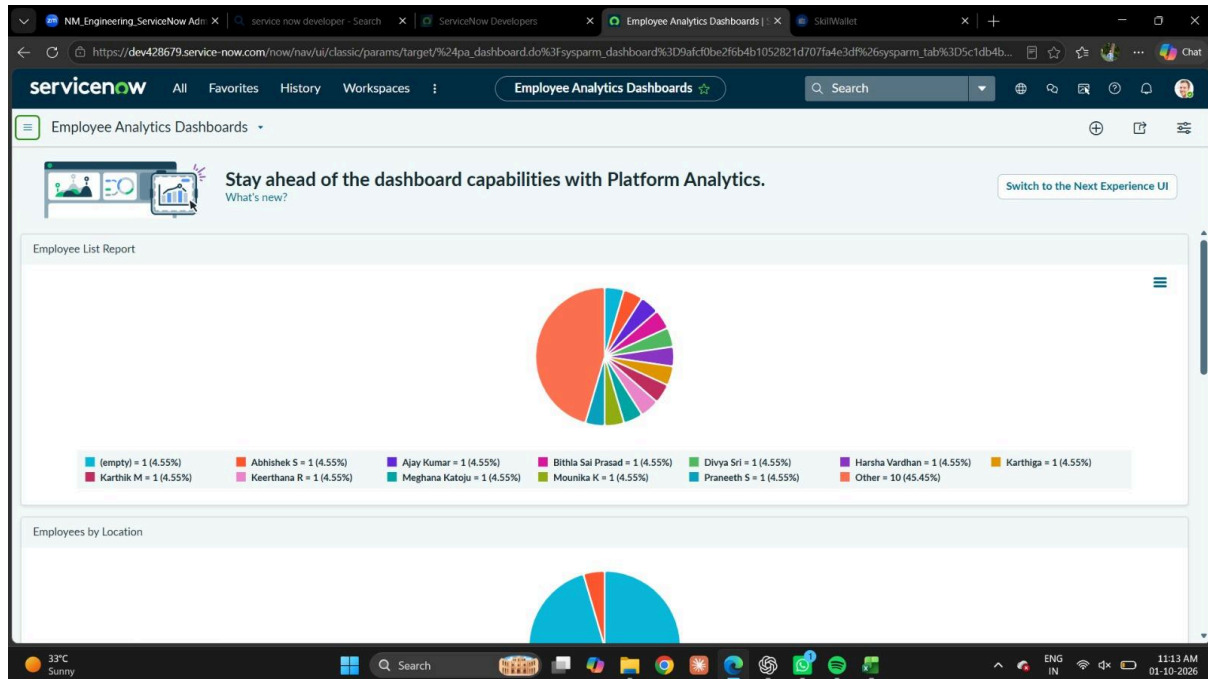
The screenshot shows the ServiceNow 'Report Stats' page. The table lists various reports with their last run times, run counts, and run times. The report 'Report: Employees by Department' is highlighted.

Report	Last run	Runs	Runs on page	Runs on schedule	Recent run time	Run time
Report: Portfolio Overview	2026-06-12 16:01:49	0	0	0	0	0
Report: System Administrator: 2015-11-10 16:15:57	2026-06-12 16:01:49	0	0	0	0	0
Report: Drilldown: System Administrator: 2015-11-12 12:20:07	2026-06-12 16:01:49	0	0	0	0	0
Report: Map Incident by caller	2026-06-12 16:01:49	0	0	0	0	0
Report: Employees by Department	2026-09-30 22:43:56	9	8	0	4,469	4,469
Report: Employees by Location	2026-09-30 22:42:28	4	3	0	4,538	4,538
Report: Employee List Report	2026-09-30 22:42:30	3	2	0	4,984	4,984
Report: test	2026-06-12 16:01:49	0	0	0	0	0
Report: Employees by Department	2026-09-28 22:53:44	1	0	0	3,805	3,805
Report: Drilldown: System Administrator: 2015-11-12 11:23:16	2026-06-12 16:01:49	0	0	0	0	0
Report: Drilldown: System Administrator: 2015-11-09 22:57:44	2026-06-12 16:01:49	0	0	0	0	0
Report: Drilldown: System Administrator: 2015-11-12 11:23:16	2026-06-12 16:01:49	0	0	0	0	0
Report: Drilldown: System Administrator: 2015-11-12 11:23:16	2026-06-12 16:01:49	0	0	0	0	0

Activity 4.2: Adding Reports to Dashboard

- Open All menus → Search for Dashboards → Under Self Service click Dashboards. Create the dashboard named Employee Analytics Dashboards.
- Open the previously created report Employees by Department. Select Edit Report, click Share, choose Add to Dashboards, select Employee Analytics Dashboards, and click Add.
- Repeat the same process for Employees by Location and Employee List Report. The reports can also be added as widgets using the + icon.
- After adding the three reports, verify that the dashboard contains the Department, Location, and List Report views. Dashboard sharing can restrict access to Users, Groups, and Roles; the background color can also be configured.

The three Reports are added successfully to the Employee Analytics Dashboards dashboard, providing a centralized view of employee data by Department, Location, and employee list.



Project Repository and Submission Structure

The project deliverables are organized according to the provided submission repository structure. The repository is intended to hold the documentation and phase-wise project artifacts.

Import-Data-Using-Transform-Maps

- 1. Brainstorming & Ideation
- 2. Requirement Analysis
- 3. Project Design Phase
- 4. Project Planning Phase
- 5. Project Development Phase
- 6. Project Testing
- 7. Project Documentation
- 8. Project Demonstration

Project Artifacts

- Data preparation: Employee Excel spreadsheet.
- Import configuration: ServiceNow Import Set Table and Transform Map.
- Data integrity: Field mapping, validation, and Employee ID Coalesce configuration.
- Target data: Employee Test records.
- Reporting: Reports and Dashboard.

Exploring the ServiceNow Project Pages

Import Set / Staging Page:

Description: The Import Set Table page represents the staging layer of the project. It is where employee spreadsheet rows are held before transformation. The page should be used to review the source values and confirm that the expected rows were loaded.

Transform Map Page:

Description: The Transform Map page defines how source employee fields are mapped to the Employee Test target table. It is also where the Employee ID field map is configured for Coalesce so matching records can be updated.

Employee Test Target Page:

Description: The Employee Test table is the destination for the transformed records. After the transform runs, this page is used to verify that the target values match the source spreadsheet and that Coalesce behavior is working as intended.

Reports / Dashboard Page:

Description: The reporting layer presents the final employee data through Employees by Department, Employees by Location, and Employee List Report. These three reports are added to the Employee Analytics Dashboards dashboard for a consolidated view of the imported employee data.

Project Results and Verification Summary

The completed workflow demonstrates the movement of employee data from Excel through ServiceNow staging and transformation into the Employee Test table, followed by reporting.

End-to-End Result

- Source: Employee data in Excel format.
- Staging: Import Set Table.
- Transformation: Transform Map with field mappings and validation.
- Data integrity: Employee ID used as the Coalesce field.
- Target: Employee Test table.

- Presentation: Reports and Dashboard.

The supplied execution results report 21 records imported successfully, duplicate behavior tested, existing records updated correctly through Coalesce, and no slow/error step observed.

Conclusion:

This project successfully implemented an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency, and data integrity throughout the employee onboarding and update process.

A key enhancement in this project was the use of Coalesce in the Transform Map. By enabling Coalesce on the Employee ID field, the system intelligently identifies existing employee records during every import. This prevents the creation of duplicate employee records and ensures that repeated imports update existing data instead of inserting redundant entries. As a result, the employee database remains clean, reliable, and consistent, which is critical in real-world enterprise environments where data may be imported multiple times from external systems.

In addition to data integrity, the project was extended with Reports and Dashboards to provide meaningful insights and real-time visibility into employee data and import activities. Custom reports were created to track:

Total employee records; Newly added employees; Updated employee records; Department-wise employee distribution; Import success and error counts.

These reports were then consolidated into a Dashboard, allowing administrators and HR teams to monitor employee data trends, import performance, and data quality from a single centralized view. Dashboards improved decision-making by presenting information visually through charts and graphs, reducing the need for manual analysis.